



Ibero-American Interuniversity Mathematics Competition (CIIM) 2024

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- 1 Let $(a_n)_{n \geq 1}$ be a sequence of real numbers. We define a sequence of real functions $(f_n)_{n \geq 0}$ such that for all $x \in \mathbb{R}$, the following holds:

$$f_0(x) = 1 \quad \text{and} \quad f_n(x) = \int_{a_n}^x f_{n-1}(t) dt \quad \text{for } n \geq 1.$$

Find all possible sequences $(a_n)_{n \geq 1}$ such that $f_n(0) = 0$ for all $n \geq 2$.

Note: It is not necessarily true that $f_1(0) = 0$.

- 2 Let n be a positive integer, and let M_n be the set of invertible matrices with integer entries and size $n \times n$.

(a) Find the largest possible value of n such that there exists a symmetric matrix $A \in M_n$ satisfying

$$\det(A^{20} + A^{24}) < 2024.$$

(b) Prove that for every n , there exists a matrix $B \in M_n$ such that

$$\det(B^{20} + B^{24}) < 2024.$$

- 3 Given a positive integer n , let $\phi(n)$ denote the number of positive integers less than or equal to n that are relatively prime to n . Find all possible positive integers k for which there exist positive integers $1 \leq a_1 < a_2 < \dots < a_k$ such that:

$$\left\lfloor \frac{\phi(a_1)}{a_1} + \frac{\phi(a_2)}{a_2} + \dots + \frac{\phi(a_k)}{a_k} \right\rfloor = 2024$$

- 4 Given the points $O = (0, 0)$ and $A = (2024, -2024)$ in the plane. For any positive integer n , Damian draws all the points with integer coordinates $B_{i,j} = (i, j)$ with $0 \leq i, j \leq n$ and calculates the area of each triangle $OAB_{i,j}$. Let $S(n)$ denote the sum of the $(n+1)^2$ areas calculated above. Find the following limit:

$$\lim_{n \rightarrow \infty} \frac{S(n)}{n^3}.$$

- 5 A board of size $3 \times N$ initially has all of its cells painted white. Let $a(N)$ be the maximum number of cells that can be painted black in such a way that no three consecutive cells (either horizontally, vertically, or diagonally) are painted black. Prove that

$$\lim_{N \rightarrow \infty} \frac{a(N)}{N}$$

exists and determine its value.

- 6 Given a real number x , define the series

$$S(x) = \sum_{n=1}^{\infty} \{n! \cdot x\},$$

where $\{s\} = s - \lfloor s \rfloor$ is the fractional part of the number s . Determine if there exists an irrational number x for which the series $S(x)$ converges.
